Certification in Agentic Systems and Design (IITRAS-2601)

Final Offline Evaluation Syllabus

Module 1: Foundations

S. No.	Lecture Title	Topics
1	**The AI Landscape for Beginners**	AI vs ML vs GenAI landscape ◆ History of AI ◆ Current industry use cases ◆ The road ahead
2	**Setting Up Your Lab**	VS Code installation ◆ Google Colab basics ◆ Managing 'Secrets' and API keys ◆ Creating a GitHub account
3	**The Developer Workflow**	Command Line (CLI) basics ◆ Git Init ◆ Add ◆ Commit ◆ The 'Notebook' vs 'Script' discipline
4	**Python: The Language of AI**	Variables ◆ Data types (Int, Float, String, Bool) ◆ Basic math in Python ◆ Print and Input functions
5	**Python: Decision Making**	Conditionals (If-Else) ◆ Boolean logic ◆ Logical operators (And, Or, Not) ◆ Indentation rules
6	**Python: Automation & Loops**	For loops ◆ While loops ◆ Iterating over simple lists ◆ Loop control (Break, Continue)
7	**Python: Organizing with Functions**	Defining functions ◆ Parameters and Arguments ◆ Return values ◆ Code modularity
8	**Python: Data Collections**	Lists and Slicing ◆ Dictionaries for key-value data ◆ Tuples ◆ Sets ◆ Choosing the right structure
9	**Python: Working with Files**	Opening files (Open, Close) ◆ Using 'With' blocks ◆ Reading TXT files ◆ Writing simple logs
10	**NumPy: Numerical Foundations**	Why NumPy? ◆ Creating arrays ◆ Basic array math ◆ Shape and Reshape ◆ Performance vs Python lists
11	**Data Structures for AI**	Multidimensional arrays ◆ Slicing NumPy ◆ Broadcast logic ◆ Generating random data
12	**Introduction to JSON**	JSON structure ◆ Dictionaries to JSON ◆ Parsing API responses ◆ Nested JSON handling

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13	**Pandas: The Data Table**	Pandas Series vs DataFrames ◆ Loading CSV files ◆ Basic inspection (Head, Tail, Info, Describe)
14	**Pandas: Selection & Filtering**	Selecting columns ◆ Row indexing ◆ Boolean filtering ◆ Sorting dataframes
15	**Pandas: Cleaning & Grouping**	Handling missing values ◆ GroupBy operations ◆ Basic aggregations (Sum, Mean, Count) ◆ Rename and Drop
16	**SQL: Thinking in Tables**	Relational vs Non-relational ◆ Primary Keys ◆ Tables and Rows ◆ Database schema basics
17	**SQL: Core Querying**	SELECT and WHERE ◆ Using DISTINCT ◆ Sorting with ORDER BY ◆ Limit and Aliases
18	**SQL: Joining Data**	Inner Joins ◆ Left Joins ◆ The Power of the JOIN ◆ SQL mental model vs Pandas merge
19	**Matplotlib: Basic Plotting**	Line plots ◆ Scatter plots ◆ Bar charts ◆ Labeling and titles ◆ Basic aesthetics
20	**Advanced Storytelling with Data**	Plotly for interactive charts ◆ Choosing the right chart ◆ Color theory basics ◆ Exporting visuals
21	**The EDA Checklist**	Standard EDA steps ◆ Identifying outliers ◆ Correlation heatmaps ◆ Building a data story
22	**APIs: The Agent's Hands**	Requests library ◆ GET vs POST ◆ Handling Status Codes ◆ API Documentation reading
23	**API Security & Ethics**	API Terms of Service (ToS) ◆ Rate limiting ◆ Environment variables for keys ◆ Data privacy basics
24	**EDA Workshop**	Ingesting raw API data ◆ Cleaning with Pandas ◆ Visualizing insights ◆ Simple Streamlit demo

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Module 2: Fundamentals of ML

S. No.	Lecture Title	Topics
25	**The ML Workflow & Habits**	Problem framing ◆ Features vs Labels ◆ Training/Validation/Test split ◆ Baselines
26	**Data Prep: Handling Messy Data**	Imputing missing values ◆ Removing duplicates ◆ Categorical encoding ◆ Scaling and Normalization
27	**Leakage & Imbalance**	Data leakage guard ◆ Class imbalance basics ◆ Synthetic data concepts ◆ Cross-validation
28	**Regression: Linear Regression Fundamentals**	Regression Basics – predicting continuous values and common real-world examples ◆ Linear Regression – intuition of best-fit line and relationship between variables ◆ Model Training – fitting regression models and generating predictions ◆ Model Complexity – understanding overfitting and generalization ◆ Residuals – interpreting prediction errors conceptually
29	**Regression: Regularization and Evaluation**	Regularization – intuition behind Ridge and Lasso regression ◆ Model Comparison – impact of regularization on model behavior ◆ Evaluation Metrics – MAE, MSE, RMSE and R-squared interpretation ◆ Error Analysis – identifying patterns in prediction errors ◆ Practical Perspective – choosing metrics and evaluating model performance
30	**Classification: Logistic Regression Fundamentals**	Classification Basics – predicting categories vs regression problems and common use cases ◆ Logistic Regression – intuition of probability based prediction and sigmoid concept ◆ Model Predictions – class predictions and probability outputs ◆ Decision Threshold – understanding probability threshold and its effect on predictions ◆ Confusion Matrix – interpreting true positives false positives true negatives and false negatives
31	**Classification Models and Evaluation Metrics**	Decision Trees – rule-based model intuition and decision paths ◆ Random Forest – ensemble concept combining multiple trees ◆ Evaluation Metrics – accuracy precision recall and F1 score interpretation ◆ ROC-AUC – understanding model ranking and separation ability ◆ Threshold Tuning – adjusting decision thresholds to balance precision and recall
32	**Unsupervised Learning: Clustering and Dimensionality Reduction**	Unsupervised Basics – learning patterns without labeled targets ◆ K-Means Clustering – grouping similar data points for segmentation ◆ K-Means Workflow – assigning points to clusters and cluster centers ◆ PCA – reducing many features into fewer components ◆ PCA Visualization – projecting high dimensional data into 2D for visualizatiUon
33	**Time Series**	Trend vs Seasonality ◆ Time-aware splits ◆ Rolling windows ◆ Evaluation for time series
34	**ML Workshop - Model Selection & Comparison**	Metric tables ◆ Comparison by complexity ◆ Model persistence (Saving/Loading) ◆ Selection checklist

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Module 3: GenAI & Agents

S. No.	Lecture Title	Topics
35	**GenAI Concepts for Beginners**	Where classical ML fails ◆ Intro to Neural Networks intuition ◆ The rise of LLMs, Tokens ◆ Context windows ◆ Hallucinations ◆ Probabilistic text generation
36	**Mastering Prompt Engineering**	System vs User prompts ◆ Zero-shot vs Few-shot ◆ Chain of Thought (CoT) ◆ Prompt templates, self-correction prompts ◆ iterative prompting ◆ agent prompt design
37	**Open-Source LLMs**	Ollama for local LLMs ◆ Local LLM demo
38	**RAG Foundations**	What is RAG? ◆ The retrieval pipeline ◆ Improving accuracy through context ◆ Simple RAG demo
39	**Embeddings & Vector Search**	Text to vectors ◆ similarity scores ◆ vector database (Chroma or FAISS) ◆ semantic search
40	**RAG: Chunking & Document Preparation**	Document loading ◆ chunk size ◆ overlap ◆ metadata basics ◆ chunk storage
41	**RAG: Retrieval & Grounded Generation**	Top-k retrieval ◆ context assembly ◆ grounded generation ◆ mini RAG app
42	**Memory & Control Flow**	short-term vs long-term memory ◆ loop termination ◆ basic error handling
43	**Agent Tool Use**	Function calling ◆ tool schemas ◆ execution cycle ◆ tool result handling
44	**LLM Internals for Designers**	Tokens vs words ◆ context limits ◆ temperature and determinism ◆ practical implications for RAG and agents
45	**Prompt Versioning & API Rate Limits**	Prompt versioning ◆ config files ◆ rate limits ◆ retries and backoff
46	**Structured Outputs for Agents**	JSON schema ◆ structured generation ◆ output parsing ◆ validation (lite)
47	**Agent Build Workshop**	Document selection ◆ ingest and chunk ◆ embed and store ◆ retrieve ◆ grounded generation ◆ simple runnable RAG app

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Module 4: Agentic Systems & Design

S. No.	Lecture Title	Topics
48	**Safety & Guardrails for Agents**	Malicious prompt injection ◆ tool safety ◆ guardrails ◆ safe code review habit (lite)
49	**Retrieval & Grounding: Chunking & Metadata**	Metadata filters ◆ chunking refresh ◆ source tagging ◆ grounding checklist
50	**Retrieval & Grounding: Hands-On Improvement**	Retrieval tuning ◆ top-k experiments ◆ failure analysis ◆ before/after comparison
51	**Multimodal Pipelines (Speech and Vision Models)**	Speech-to-Text ◆ Transcript summarization ◆ Text-to-Speech integration ◆ Vision basics
52	**Agent Communication Patterns**	Planner–executor ◆ task decomposition ◆ JSON message contracts ◆ stop conditions
53	**LangGraph Basics**	Nodes and transitions ◆ graph state ◆ simple graph build ◆ execution walkthrough
54	**LangGraph Advanced: Checkpoints & Resume**	Checkpoints ◆ persisted state ◆ resume workflow ◆ inspect checkpoint data
55	**LangGraph Advanced: Timeouts & Retries**	Timeouts ◆ retry policy ◆ user-facing errors ◆ reliability checklist (lite)
56	**Observability & Tracing for Agents**	Step tracing ◆ structured logs ◆ log fields ◆ read-only debug workflow
57	**LLMOps: Evaluation Frameworks**	Golden task set ◆ offline eval run ◆ qualitative scoring ◆ regression habit
58	**LLM Operations: Versioning, Eval Gates & Cost**	Release versioning ◆ pre-release eval gate ◆ token and cost tracking ◆ secrets handling (lite)
59	**Deployment: Streamlit User Interface**	Streamlit layout ◆ user input ◆ display agent trace (lite) ◆ local run
60	**Deployment: API Facade & Hosting Basics**	FastAPI facade (lite) ◆ request/response contract ◆ hosting options ◆ environment config
61	**Ops: Caching & Concurrency**	Response caching ◆ rate-limiting users ◆ queue awareness (lite) ◆ cost awareness

Note:
All the topics covered in the live sessions will be considered as part of the evaluation syllabus.